

Class 1 — Alterations in Cell Function & Growth, Part 1 (The Cell, part 1)

NRS224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture (S# = slide, mm:ss = time in the recording), the in-class slides, and your full study sheet pp. 1-4.

Page 1 — The cell: components & functions

Guide prompt: "If pathological changes occur in any of these components → Nursing problems:"

1.1 → problems with nucleus:

- Nucleus = control center → DNA holds instructions for protein synthesis; RNA carries them to cytoplasm [S3-S4, 02:54-06:32; sheet p.1]
- Damaged nucleus → instructions get messed up → can't make certain proteins at all, OR proteins made but defective → don't work [S4, 04:08-06:32]
- Bottom line: trouble with protein synthesis [S4, 04:08-06:32; sheet p.1]

1.2 → problems with ribosomes:

- Ribosomes make proteins (site of protein synthesis out in cytoplasm)
[S6, 07:01-07:34; sheet p.1]
- Damaged ribosomes → same problem: proteins not made, or made defective and won't work [S6, 07:01-07:34]
- Class practice Q: "If the function of ribosomes were impaired..." → best answer: "Synthesis of defective proteins, or decreased protein synthesis" [class slide 6]

1.3 → problems with ER:

- ER = system of tubules: (1) transports substances from place to place inside cell, (2) houses ribosomes + enzyme systems (enzymes = catalysts that speed up chemical reactions) [S8, 07:54-09:42; sheet p.1]
- Problems: can't get things place to place; ribosomes have nowhere to hang out → protein synthesis problems; chemical reactions slow way down [S8, 07:54-09:42]
- Class practice Q: impaired ER → best answer: "Substances may not be transported appropriately within the cell" [class slide 7]

1.4 → problems with mitochondria:

- Mitochondria = powerhouse ("Mighty Chondrion") → transforms food energy into ATP; needs oxygen to do it
[S11, 10:28-12:06; sheet p.1]
- Not enough O₂ → can't make ATP → power failure → cell can't carry out activities → dies

[S11, 10:28-12:06]

- Her examples: heart attack = heart muscle cells die from power failure; stroke = brain cells [S11, 10:28-12:06]

1.5 → problems with lysosomes (and peroxisomes):

- Lysosomes = bags of digestive enzymes → get rid of worn-out cell parts / foreign material (e.g., bacteria) [S12, 12:06-12:33; sheet p.1]
- Not working → things accumulate inside cell = lysosomal storage disease → cell damaged or malfunctions [S13-S14, 12:33-13:08]
- Example: Tay-Sachs (missing lysosomal enzyme → accumulation in retina + nervous system → blindness, motor/mental issues) [S15, 13:08-14:15]
- Peroxisomes = specialized lysosomes that detoxify/get rid of free radicals (unstable atom/molecule that can damage cells) → free radical damage blamed for wrinkles, cancer, atherosclerosis, heart disease [S19, 15:16-16:36; sheet p.1]
- ★ **Prof emphasis:** she won't test on Tay-Sachs specifics — "more likely to ask" what happens if lysosomes aren't working [S18, 14:43-15:16]

1.6 → problems with Golgi complex:

- Golgi = post office of the cell → takes ribosome-made proteins, packages them into secretory vesicles/granules, directs them to the right place [S20-S21, 16:36-17:55; sheet p.1]

- Not working → proteins made don't get to the right place → can't do their job [S21, 17:06-17:55]
- Example: I-cell disease — lysosome enzyme packaged/released into blood → lysosome missing enzyme (storage disease) + enzyme breaks down things in blood you still want → damage to heart, lungs, nervous system, MSK deformities [S22-S23, 17:55-20:36]
- ★ **Prof emphasis:** no test questions specifically on I-cell disease — only the underlying pathology of lysosome/Golgi malfunction [S23, 19:23-20:36]

1.7 → problems with cell membrane:

- Membrane holds cell together — keeps what belongs inside in, keeps out what doesn't; carrier/transport proteins + ion channels let O₂/fuel in and waste out [S24, 20:36-21:35; sheet p.1]
- Receptor sites (hormones/chemical messengers) → cellular communication; damaged receptors → poor communication AND medications that stimulate/block receptors won't be effective [S26, 22:11-24:26]
- Self-antigens tell immune system "self vs non-self" → if damaged, immune system may attack self-cells = autoimmune disease [S26, 22:11-24:26]
- Also maintains resting membrane potential + generates action potentials; worst case: cell breaks apart and dies [S26, 22:11-24:26]

Page 2 — Cell metabolism & movement across the membrane

2.1 Nursing problem (altered cellular metabolism/energy production):

- Cellular metabolism = cell makes energy: food → ATP, stored until needed [S27, 24:26-25:26]
- Two pathways: aerobic (requires O₂, done in mitochondria) vs anaerobic (no O₂, less efficient, fewer ATP, but a good backup) [S27-S29, 24:26-26:26; sheet p.2]
- Anaerobic too long → lactic acid → changes body fluid pH = acidosis → lots of things don't function normally [S30, 26:26-27:53]
- 3 things cell needs for energy: fuel, oxygen, appropriate enzymes (catalysts) → missing any one → not enough energy → cells malfunction or die [S30, 26:26-27:53; sheet p.2]

2.2 Pathological change in movement across membrane → Nursing problem:

- 3 ways across membrane: passive movement, active transport, endocytosis/exocytosis [S31, 27:53-29:00; sheet p.2]
- Passive = down concentration gradient (high → low), no energy, goal = equilibrium; diffusion = particles move, osmosis = water moves; deciding factor = membrane permeability [S32-S43, 29:00-34:32]
- Hypernatremia example: too much Na⁺ in blood → bigger gradient → Na⁺ still can't cross → MORE water osmoses out → cell shrinks →

brain cells shrink = lots of neuro issues

[S44-S47, 34:32-37:04]

- Active transport = against gradient, requires energy (ATP) — Na^+/K^+ pump keeps K^+ in, Na^+ out → needed for action potentials
[S48-S49, 37:04-40:38]
- Endocytosis (pinocytosis = fluid/dissolved "cell drinking"; phagocytosis = solids, e.g., neutrophils/monocyte-macrophages eating bacteria) → problem = big impact on immunity; exocytosis releases hormones/gets rid of waste → problem = can't release hormones [S50, 40:38-42:58]

Page 3 — Cellular communication & membrane potentials

3.1 Nursing problem (alterations in availability/function of chemical messengers or receptor sites):

- Chemical messenger method: one cell releases hormone/neurotransmitter/growth factor → binds receptor site on 2nd cell → tells that cell what to do [S51, 42:58-46:13; sheet p.2]
- Two places it breaks: (1) too much or too little messenger, (2) damaged or missing receptor sites → message never gets delivered
[S51, 42:58-46:13]
- Her examples: Parkinson's = not enough dopamine (tremor, slow stiff movements); schizophrenia = too much dopamine; myasthenia gravis = autoimmune attack on receptors at neuromuscular junction → muscle weakness [S51-S52, 42:58-47:34]

- Predisposing factors per sheet: autoimmune dz, congenital defects, genetic predisposition, degeneration d/t age or toxin exposure

[sheet p.2]

- Meds work by increasing/decreasing messenger or stimulating/blocking receptors

[S52, 46:13-47:34; sheet p.2]

3.2 Nursing problem (changes in membrane permeability):

- Resting membrane potential: cell electrically polarized, inside more negative (K^+ diffuses out, Na^+ can't get in → mostly anions left inside) [S54-S58, 49:37-52:59; sheet p.3]
- Action potential: something changes membrane permeability → Na^+ diffuses in → inside becomes positive = depolarization; ions then pass cell-to-cell through gap junctions (nerve/muscle communication)

[S59-S61, 52:59-55:25]

- Repolarize (2 steps): membrane goes back to not letting Na^+ in + Na^+/K^+ pump pumps K^+ back in, Na^+ back out → ready to repeat

[S62-S64, 55:25-57:00]

- Problems: electrolyte imbalance messes up the gradient; hypoxia/impaired metabolism (no O_2 or glucose) → no ATP → no Na^+/K^+ pump → can't repolarize → brain & heart can't function [S66, 57:19-59:30; sheet p.3]

Page 4 — Cell injury: adaptive responses, reversible injury, cell death

Guide lists each response with space under it. Fill: what it means clinically, upside/downside, and her example(s).

4.1 atrophy

- Cells get smaller / revert to lower level of functioning when not enough O₂, fuel, or blood flow → smaller cell needs less O₂/nutrients → more likely to survive [S79, 74:33-76:03; sheet p.4]
- Upside: cell survives, doesn't die. Downside: not as functional [S79, 74:33-76:03]
- Her examples: calf muscle atrophy from chronic impaired arterial blood flow → can't do as much work; atrophied endocrine gland → less hormone [S80, 76:03-76:47]

4.2 hypertrophy

- Cells get bigger in response to increased work demands — opposite of atrophy [S81, 76:47-77:05; sheet p.4]
- Examples: bodybuilders lifting heavier weights (skeletal muscle cells); left ventricle in chronic high BP (pushing against high arterial pressure → heart muscle cells hypertrophy) [S82-S83, 77:05-77:56]
- Upside: cells can do more work. Downside: bigger cells doing more work need more blood, O₂, fuel → if atherosclerosis limits supply → may set person up for a heart attack [S84, 77:56-78:40]

4.3 hyperplasia

- Increase in cell NUMBER (not size) — usually in response to chemical stimulus: change in hormone or growth factor
[S85, 78:40–80:31; sheet p.4]
- Examples: wound healing (growth factor → more cells to fill in wound); pregnancy uterus = BOTH hypertrophy + hyperplasia
[S85, 78:40–80:31]
- Downside: tissue enlarges enough to interfere with surrounding structures — benign prostatic hyperplasia squeezes urethra shut → trouble emptying bladder [S86, 80:31–81:29]
- ★ **Prof emphasis:** "students do mix those two terms up a lot on the test" — hypertrophy = size, hyperplasia = number [S85, 78:40–80:31]

4.4 metaplasia

- Cells change from one cell type to another that's better able to survive — triggered by chronic irritation/inflammation
[S87, 81:29–82:19; sheet p.4]
- Not a change out of primary type (not nerve→muscle); e.g., one kind of epithelial cell → a sturdier one [S87, 81:29–82:19]
- Her example: smoker's airway ciliated columnar epithelium → stratified squamous (sturdier, less likely to be injured)
[S88, 82:19–83:14]
- Downside: lose protective cilia → mucus + trapped bacteria/irritants get stuck in airway, can't get out [S88, 82:19–83:14]

4.5 dysplasia

- Book's definition she likes: "deranged cell growth" — cells can be all different shapes and sizes, don't really look like each other
[S90, 83:45-84:27; sheet p.4]
- Usually a reaction to chronic irritation/inflammation; progression: normal → metaplasia → dysplasia if irritation isn't removed [S90-S91, 83:45-86:00]
- No upside; big downside = precursor to cancer (higher risk, NOT a guarantee)
[S91, 84:27-86:00]
- Examples: airways of smokers; uterine cervix — Pap smear finding dysplastic cells ≠ cancer, means higher risk → more frequent Paps
[S91, 84:27-86:00]

4.6 reversible injury

- Cellular response ladder depends on severity of stressor: adapt → become injured → die
[S78, 74:10-74:33; sheet p.3]
- If the stress is severe the cell becomes injured anyway; if we get rid of the source of the stress, cell may heal and injury be reversed
[S92, 86:00-89:15]
- If injury is bad enough, the cell will actually die (→ 4.7) [S92, 86:00-89:15]

4.7 cell death/necrosis - if cell injury is irreversible

- How bad depends on how many cells die: a few liver cells = probably fine; a whole bunch

= liver failure → patient may die

[S92, 86:00-89:15]

- Replacement varies: skin cells replace themselves all the time; brain cells do so at a much, much slower rate [S92, 86:00-89:15]
- Dead cells replaced either by same cell type OR by scar tissue — heart attack: heart muscle cells replaced with scar tissue → restores structure but NOT function (no action potentials, no contraction) → chronic heart failure [S92, 86:00-89:15]
- Lab markers of cell damage: draw blood, look for proteins/enzymes normally found only inside cells → high levels = cells damaged, insides leaking out. Troponin → heart muscle cells (chest pain in ED) [S74, 69:15-70:51]
- CK and LD → "damage to the liver or the skeletal muscles or some tissues like that" — she does not pair them one-to-one (your sheet writes CK → skeletal muscle, LD → liver)

[S74, 69:15-70:51; sheet p.3]